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Low rate of return to pre-injury sport level in athletes after cartilage surgery: a 10-year follow-up study

S. Zaffagnini[1] · F. Vannini[2] · A. Di Martino[1] · L. Andriolo[1] · A. Sessa[1] · F. Perdisa[1] · F. Balboni[3] · G. Filardo[4] · The ESSKA U45 Committee

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Abstract

Purpose Although articular surface is frequently damaged in athletes, results in terms of return to sport and level of activity after cartilage surgery remain rather unpredictable and poorly documented. The aim of this study is to evaluate the clinical outcome in terms of rate and level of return to sport in a group of competitive athletes who underwent matrix-assisted autologous chondrocyte transplantation (MACT), as well as the impact on their athletic career.

Methods Thirty-one male patients (mean age 22.6 ± 6.3 years) p racticing sport at competitive level, affected by focal chondral/osteochondral lesions of the distal femur, were enrolled and treated with arthroscopic hyaluronan-based MACT. Patients were evaluated prospectively at 1-year intervals with the IKDC subjective, Tegner, and EuroQol VAS scores during their pre-operative visit and subsequent follow-ups for up to 10 years. Return to sport in terms of level, time and maintenance of the activity level was documented, together with surgical or clinical failures.

Results A marked improvement in all scores was found: IKDC increased from 40.3 ± 13.4 to 81.7 ± 14.4 ($p < 0.0005$) at 12 months; a further improvement was observed at 2 years (89.5 ± 11.3; $p = 0.008$), then results were stable for up to 10 years (87.3 ± 13.6). The analysis of return to sport documented that 64.5% of patients were able to return at a competitive level, and 58.1% performed at the same pre-injury level, with activity rates decreasing over time. The rate of patients returning to competitive level was 84% in those without previous surgery (vs. 33% who had undergone previous surgery), 87% for those with traumatic lesions (vs. 33% and 50% for degenerative and OCD lesions, respectively), and 92.3% in younger patients (age < 20 years). Among these factors, multivariate analysis demonstrated that previous surgery was the single most influencing factor for returning to the same sport level ($p = 0.010$).

Conclusions These long-term results showed that chondrocyte-based regenerative approach has some limitations in terms of sport-related outcomes. The level of high functional knee restoration needed for such high-demanding activity level can be challenging to achieve, especially in patients with a more compromised joint homeostasis. Return to sport rate varies significantly according to specific patient and lesion characteristics and best results are obtained in young patients with traumatic lesions without previous surgery, which should be considered when treating athletes affected by cartilage lesions.

Level of evidence IV.

Keywords Knee · Cartilage · Chondrocyte · Scaffold · Regeneration · Sport

✉ L. Andriolo
Lucas.andriolo@gmail.com

1 Clinica Ortopedica e Traumatologica II, IRCCS Istituto Ortopedico Rizzoli, Via di Barbiano 1/10, 40136 Bologna, Italy

2 Clinica Ortopedica e Traumatologica I, IRCCS Istituto Ortopedico Rizzoli, Bologna, Italy

3 Laboratorio NABI, IRCCS Istituto Ortopedico Rizzoli, Bologna, Italy

4 Applied and Translational Research (ATR) Center, IRCCS Istituto Ortopedico Rizzoli, Bologna, Italy

Introduction

The articular surface is frequently damaged in athletes, with prevalence reported to be up to 89% [12, 32]. These injuries can lead to debilitating symptoms and, if left untreated, may act in combination with repetitive joint loading during sports activity, eventually favoring the development of early degenerative changes in the affected joint [5, 16]. In particular, knee injuries account for 46% of career-ending injuries in professional soccer with over a quarter resulting

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from cartilage injuries; moreover, athletes are more likely to develop osteoarthritis than the general population [12, 29, 30].

Among traditional surgical options to address articular cartilage defects in athletes, microfractures are documented to provide good results when applied to small lesions, with a fast return to sport [21]. Although a short rehabilitation time is ideal for athletes [22], microfractures only restore a fibrocartilaginous tissue with limited durability, and a time-dependent deterioration of the outcome has been reported [16, 17]. The more ambitious regenerative procedures, such as autologous chondrocyte implantation (ACI) and the following generation of matrix-assisted autologous chondrocyte transplantation (MACT), restore a hyaline-like tissue, leading to satisfactory outcome over time in the general population [7, 8, 25, 27]. The use of chondrocyte-based procedures remains debated in athletes, due to the longer rehabilitation time compared with microfractures [16, 19], but MACT was shown to provide better subjective results at mid-term follow-up in professional soccer players, compared with microfractures [16], and accelerated rehabilitation was reported to be a successful approach [4]. Nevertheless, current evidence is still very poor about results of regenerative procedures in athletes, and long-term results are lacking.

Another limitation of the current literature regards the evaluation methods of sport activity. In fact, although the overall results of published studies show the possibility to return to high-demand sport activity after cartilage surgery [19], to date only a few studies specifically focused on the rate of Return To Sport (RTS) and the level of activity after treatment [13], and these aspects remain rather unpredictable and poorly documented [17, 22].

Actually, returning to sport activity at pre-injury level is the main expectation for competitive athletes undergoing such an important decision, which may require more than 1 year away from competitions. Thus, it is important to understand the potential of cartilage treatments and have the right expectations in terms of level and durability of their results.

The hypothesis was that cartilage lesions may have a negative impact on athletic career over time, despite surgical treatment. The aim of this study is to evaluate the clinical outcome in terms of rate and level of RTS at long-term follow-up in a group of competitive athletes who underwent MACT.

Materials and methods

Arthroscopic MACT was indicated for patients with focal grade III–IV (International Cartilage Repair Society—ICRS evaluation package [14]) symptomatic chondral knee defects involving femoral condyles and trochleae. Contraindications were: untreated misalignment or instability, osteoarthritis or bipolar ("kissing") lesions, and those with other general medical conditions (diabetes, rheumatoid arthritis, etc.). Patients who presented with an axial deviation or an anterior cruciate ligament (ACL) lesion underwent the combined surgical procedure of realignment or ligament reconstruction in the same surgical session as cell-loaded scaffold implantation.

From 2000 to 2006, 34 male patients practicing sport at competitive level (Tegner level 8, 9, 10), affected by focal osteochondral lesions of the distal femur, were enrolled and treated with arthroscopic MACT. Three were lost at follow-up, while 31 patients were prospectively followed at 1-year intervals for up to 10 years. Of these patients, 26 were involved in soccer participation, 2 ski, 1 basketball, 1 judo, and 1 athletics. Fourteen patients were professional athletes, nine semi-professional, and eight played at competitive level in lower categories. Patients had a mean age of 22.6 ± 6.3 years and a mean BMI of 24.1 ± 2.0. Lesions had a mean size of 2.1 ± 0.7 cm^2 and were located at the medial femoral condyle (MFC) in 18 patients, at the lateral femoral condyle (LFC) in 8, and in 2 cases the trochlea was involved. In three patients, the lesion was double, involving MFC and trochlea in two cases and LFC and trochlea in the other one. Twelve patients had undergone at least one previous surgery (38.7%), for a total of 18 previous procedures: 6 arthroscopic debridement, 5 arthroscopic meniscectomy, 3 ACL reconstruction, 2 mosaicplasty, and 2 microfractures; among these, 4 patients (12.9%) had been previously treated with other cartilage repair techniques with unsuccessful results and recurrence of symptomatology. In 15 patients, the lesion had a clear post-traumatic etiology, whereas 10 patients presented an osteochondritis dissecans (OCD) defect, and in 6 patients the etiology was micro-traumatic/degenerative.

Surgical technique and post-operative protocol

The procedure was performed in two surgical steps. The first one consisted in a 150–200-mg healthy cartilage biopsy from a non-weight-bearing knee area for chondrocyte cell culture and subsequent seeding onto the hyaluronan-based scaffold: Hyaff 11® (Fidia Advanced Biopolymers Laboratories, Padova, Italy). The second one consisted in implanting the bioengineered tissue Hyalograft C® (Fidia Advanced Biopolymers Laboratories, Padova, Italy) by an arthroscopic approach [20]. In case of OCD, due to concurrent subchondral bone loss, an autologous cancellous bone graft, harvested from the ipsilateral tibia, was used, during the first arthroscopic step, to fill the bottom of the lesion [9].

In the early stage (0–6 weeks), rehabilitation was focused on controlling pain, effusion, loss of motion, and muscle atrophy, and to protect the transplant. Self-assisted mobilization of the knee or continued passive motion, controlled mobilization exercises with reduced range of motion, and

early isometric and isotonic exercises were performed. Full weight bearing was usually allowed within 6–8 weeks after surgery, followed by functional training aimed at returning to a correct running pathway by proprioceptive, strength and endurance exercises, and aerobic training. However, rehabilitation was tailored according to combined surgeries and specific patients and sports requirements.

Patient's evaluation

A pre-operative clinical and physical examination was completed on each patient. History of possible knee trauma and previous treatments was investigated. Pre-operatively, patients were evaluated for instability, malalignment, and range of motion. Patients were then evaluated prospectively at 1-year intervals for up to 10 years of follow-up. All patients were asked to answer IKDC subjective evaluation of knee symptoms and functional and activity level, Tegner sports activity level and EuroQol VAS during their pre-op visit and subsequent follow-ups. The operation was deemed to have failed if the patient needed reoperation because of symptoms due to primary defects. For failed patients, the last clinical assessment before reoperation was considered for final evaluation. Besides surgical failures, patients without a clinically significant improvement (10 IKDC subj points as per literature definition [6]) with respect to the basal evaluation were considered clinical failures. The Hospital Ethics Committee and Internal Review Board of the Rizzoli Orthopaedic Institute, Bologna, Italy (prot. gen. n. 39667) approved this study, and informed consent of all patients was obtained.

Statistical analysis

All continuous data were expressed in terms of the mean and the standard deviation of the mean, for Tegner score data are also expressed in terms of the median and the range, the categorical data were expressed as frequency and percentages. The Kolmogorov–Smirnov test was performed to test normality of continuous variables. The Repeated Measures General Linear Model (GLM) with Sidak test for multiple comparisons was performed to assess the differences at different follow-up times. The Friedman non-parametric test, followed by the Wilcoxon post hoc pairwise comparison corrected by Bonferroni method for multiple comparisons, was used to measure differences at different follow-up times of not normally distributed scores. The ANOVA test was performed to assess the between groups differences of continuous, normally distributed and homoscedastic data, the Mann–Whitney test was used otherwise. The ANOVA test followed by the Scheffè post hoc pairwise comparison was also used to assess the among groups differences of continuous, normally distributed and homoscedastic data, the Kruskal–Wallis test followed by the Mann–Whitney test with the Bonferroni correction for multiple comparison was used in all other cases. The Spearman rank correlation was used to assess the correlation between continuous data. The Kendall tau correlation was used to assess the correlation between ordinal data. The Logistic Regression with the Wald backward method was used as multivariate analysis to assess those factors influencing the return to sport at competitive level. For all tests, $p < 0.05$ was considered significant.

All statistical analyses were performed using SPSS v.19.0 (IBM Corp., Armonk, NY, USA).

Results

No complications related to the implant or severe adverse events were observed during the treatment and follow-up period. A marked improvement in all scores was found after treatment. Only one failure was recorded in this survey, a 19-year-old patient with traumatic lesion who underwent further surgery after 6 years (transplant debridement and microfractures, and medial meniscal allograft transplantation).

The IKDC subjective score increased from a basal level of 40.3 ± 13.4 to 81.7 ± 14.4 ($p < 0.0005$) at 12 months; a further improvement was observed at 2-year follow-up reaching a score of 89.5 ± 11.3 ($p = 0.008$), then results were stable over time up to the 10-year follow-up with a level of 87.3 ± 13.6 (Fig. 1). A statistically significant improvement was observed in the EQ-VAS score from basal evaluation to 12-month follow-up (56.6 ± 19.8 and 81.8 ± 11.4, respectively, $p < 0.0005$), with a further improvement at 36-month follow-up (90.2 ± 8.8, $p = 0.01$), then results remained stable with a 10-year value of 91.1 ± 9.9 (Fig. 2). The Tegner score showed that sport activity level in this athletic population had a statistically significant improvement from pre-treatment levels [mean 1.9 ± 1.6, median 1 (range 1–7)] to all follow-up times ($p < 0.0005$). The level at 12 months [mean 6.6 ± 2.2, median 7 (range 2–10)] further improved at 2 years [mean 7.9 ± 1.8, median 9 (range 3–10); $p = 0.011$]. Afterwards, the score remained stable up to the 10-year evaluation [with a slight progressive decrease to mean 6.7 ± 2.3, median 7 (range 2–10), n.s.]. Despite the marked post-operative improvement, the mean activity level remained lower compared to pre-injury [mean 9.1 ± 2.3, median 9 (range 8–10)] ($p < 0.0005$, Fig. 3).

The analysis of factors influencing clinical results documented an influence of etiology on the IKDC subjective score, with traumatic lesions maintaining high scores during all follow-up times (93.9 ± 6.7, 92.3 ± 12.9, and 88.2 ± 13.4, at 2-, 5-, and 10-year follow-up, respectively), degenerative lesions showing a marked decrease at long-term (93.3 ± 6.4, 93.7 ± 6.2, and 79.3 ± 17.0, at 2-, 5-, and 10-year follow-up,

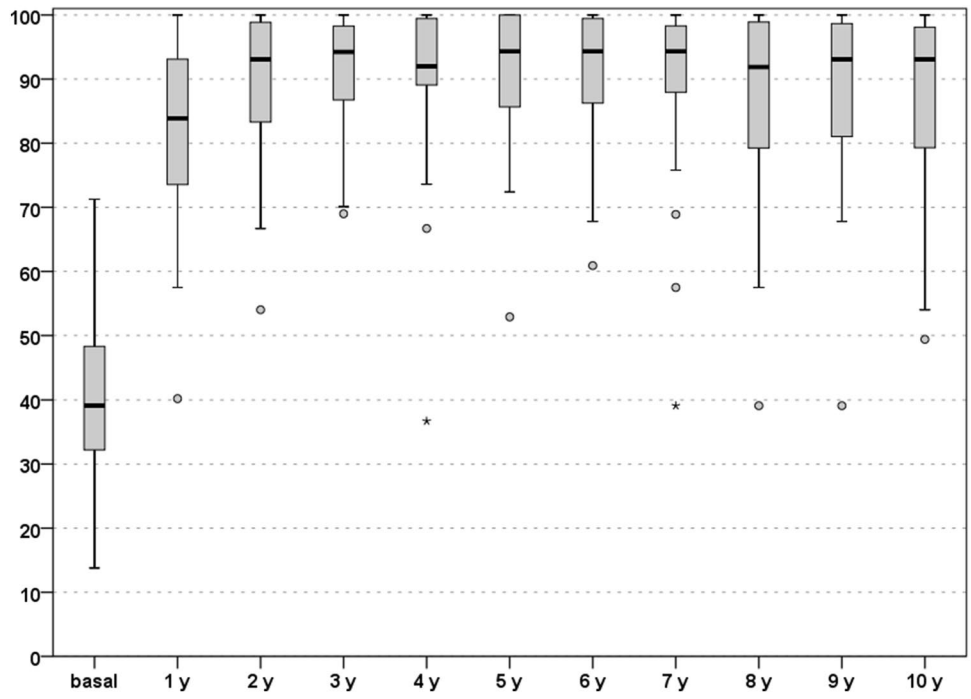

Fig. 1 IKDC subjective score improvement from basal up to 10-year follow-up

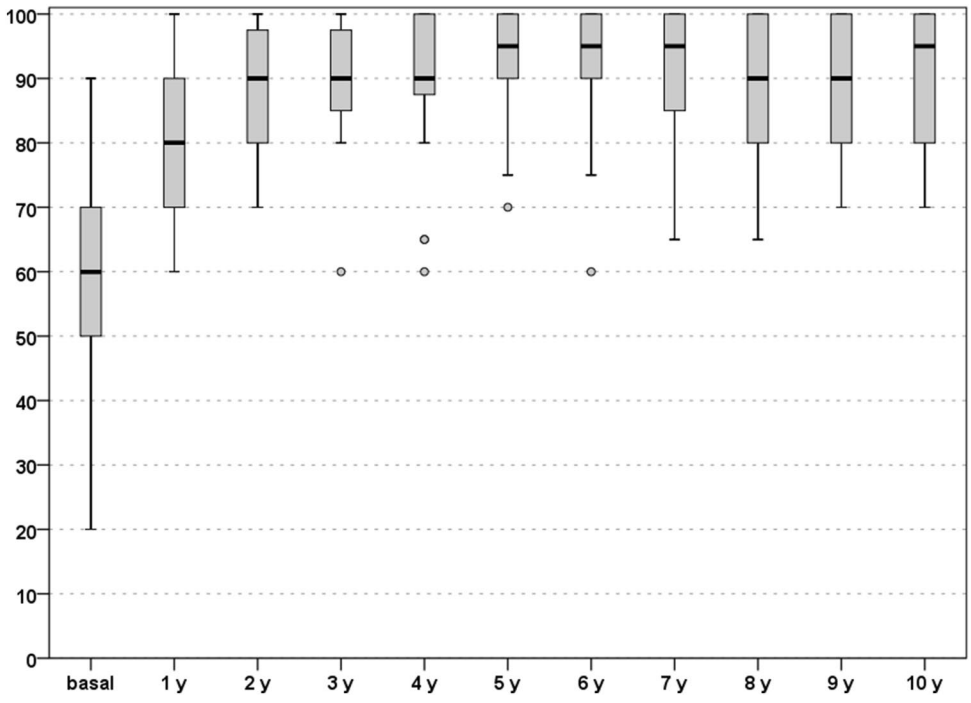

Fig. 2 EQ-VAS score improvement from basal up to 10-year follow-up

respectively), and OCD lesions presenting overall high scores but a slower improvement (80.7 ± 14.2, 88.4 ± 9.6, and 90.8 ± 11.2), at 2-, 5-, and 10-year follow-up, respectively, ($p = 0.027$; ETA2 = 0.225) (Fig. 4). With regard to activity level, higher age and BMI were found to be correlated with a lower 2-year Tegner score (both $p = 0.004$). Finally, previous surgery presented a prognostic role ($p = 0.02$), with patients without previous surgery returning to the same pre-injury Tegner level at 2 years (pre-injury 9.2 ± 0.5, 2 year 8.7 ± 1.1, n.s., Fig. 5).

The analysis of return to sport documented that 64.5% of patients were able to return at a competitive level (Tegner 8, 9, or 10) during the study follow-ups, with a 58.1% rate of same pre-injury level. When looking at the specific

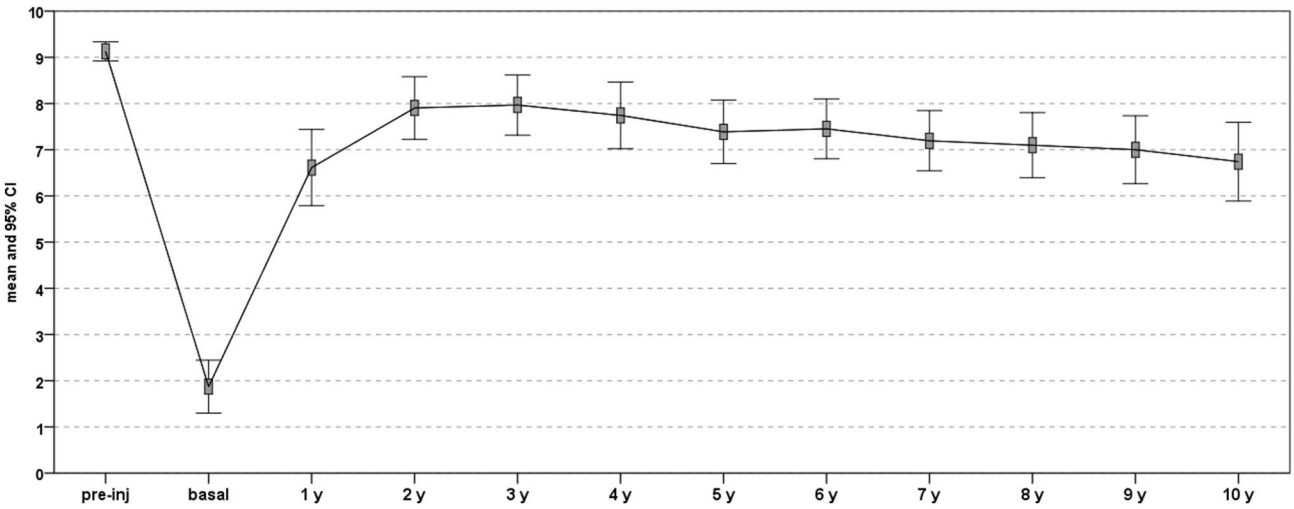

Fig. 3 Tegner activity level scale improvement from basal up to 10-year follow-up

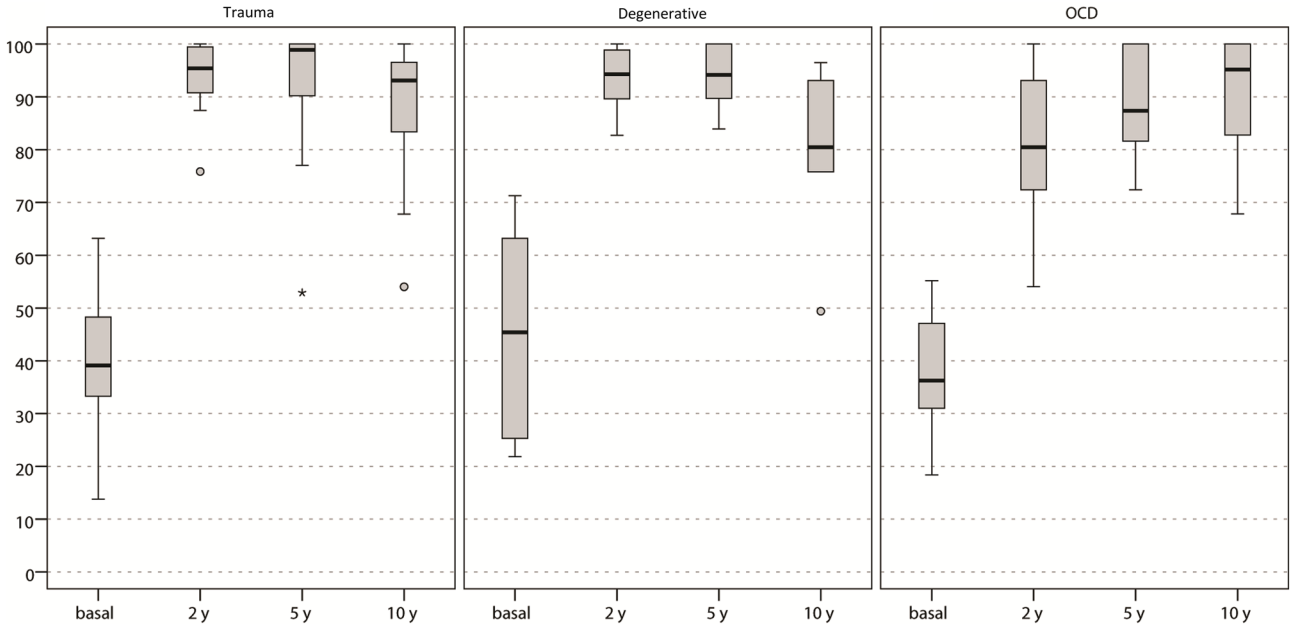

Fig. 4 Etiology influenced significantly the IKDC subjective score, with highest results for traumatic lesions

follow-up times, both rates decreased over time, with 61.3%, 45.2%, and 35.5% of patients performing at competitive level at 2, 5, and 10 years, respectively, and 54.8%, 38.7%, and 29.0% of patients performing at the same pre-injury level at 2, 5, and 10 years, respectively. The analysis of factors influencing return to sport showed that the rate of patients returned to competitive level (Tegner 8, 9, and 10) was 84% in those without previous surgery, whereas only 33% of patients who had undergone previous surgery did ($p = 0.007$). Similarly, etiology influenced this rate: patients with traumatic lesions obtained better results than patients with degenerative or OCD lesions (87% of return to competitive level vs 33% and 50%, respectively; $p = 0.035$). In addition, patient age represented a prognostic factor ($p < 0.0005$): younger athletes (< 20 years) presented 92.3% and 84.6% of return to sport rates at competitive level and at the pre-injury level, respectively. Finally, a higher BMI was correlated to a lower rate of return to sport at competitive level ($p = 0.001$). Among these factors, a multivariate analysis demonstrated that previous surgery [$p = 0.029$, $\mathrm{Exp}(B) = 139.7$] and BMI [$p = 0.032$, $\mathrm{Exp}(B) = 0.203$] were the most influencing factors on the rate of return to competitive level, and the same

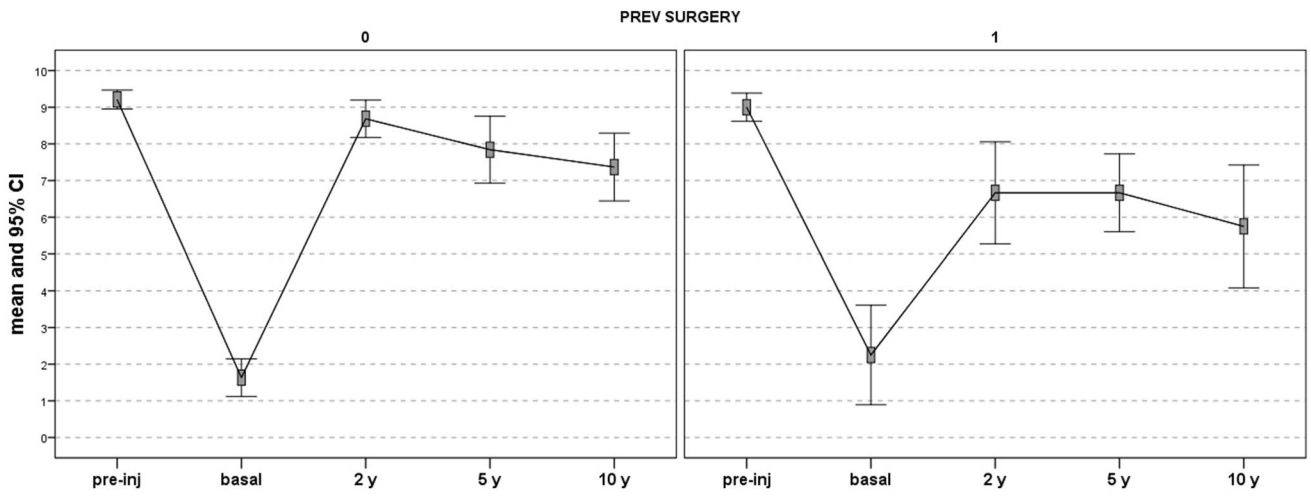

Fig. 5 Previous surgery was an influencing factor on the activity level, with significantly lower Tegner scores at all follow-ups

was found for the rate of return at the same sport level [previous surgery: $p = 0.010$, Exp(B) = 25.13; BMI: $p = 0.016$, Exp(B) = 0.421].

Discussion

The main finding of the present study is that MACT led to a satisfactory clinical subjective outcome in a group of competitive athletes, who showed stable high clinical scores in the long-term follow-up. Nevertheless, only about 60% of these athletes reached the highest goal of returning to sport at the same pre-injury level, but this rate varied significantly according to specific patient and lesion characteristics.

Athletes may represent an ideal target for regenerative cartilage procedures, since previous studies have shown that an active lifestyle and physical training can improve the benefits of ACI [4, 18, 33]. Moreover, these patients are usually highly motivated to strictly follow the rehabilitation protocols to recover optimal knee function. On the other hand, treating articular cartilage in joints under repetitive high-level stress is particularly challenging, the high functional level requested by these patients is hard to reach and, consequently, so is the possibility to return to sport.

The study results demonstrated high subjective scores and a high activity level already at short-term follow-up, which remained stable for 10 years after surgery. The mean Tegner scores at the short-mid- and long-term follow-ups document the return to sport practice, which may be accounted as a success for the treatment. The slight, progressive, not statistically significant, decrease of the outcome observed over time may be compatible with physiological aging of the study population, which could not maintain an elite activity level in high-impact sports while entering the middle age for reasons not just related to the joint health. However, although the overall return to sport activity is a successful outcome, it is important to underline that pre-injury level was not reached at any follow-up times. This was confirmed by the rate of return to sport at the same pre-injury level: 2 years after surgery only about 55% of patients played sport at the same pre-injury level, and this rate further decreased at 5 and 10 years of follow-up. This represents a key finding of this study, since the possibility of returning to the same level is paramount in this type of competitive sport active population when deciding if undergoing this surgical treatment [1]. Ardern et al. [2] proposed the use of a specific participation measure such as the ability to return to pre-injury level to better identify responder patients and guide treatment decision. The level of activity is of much greater importance for athletes than any score or functional parameters [31]. In this light, the ratio of patients able to return to the same level of sport activity has to be considered a main outcome.

Previous literature mostly fails to document this key aspect, reporting an overall return to sport while overlooking the investigation of the specific activity level achieved. Mithoefer et al. [21] reported an overall 73% of athletes being successfully back to high-impact sports participation after cartilage surgery. A meta-analysis involving 2549 athletes found that cartilage restoration surgery had a 76% return to sport rate at mid-term follow-up [12], whereas Campbell et al. focused on ACI procedures showing that 84% of athletes returned to sport participation, with smaller defects, younger age, and no previous cartilage treatment being positively correlated with the clinical result [3]. However, there is a lack of data focusing on the return to sport, documenting the level achieved, and maintained over time, following cartilage procedures in surveys composed only by elite competitive athletes. The population included in the present study had a pre-injury Tegner score ranging from 8

to 10: although the overall return to sport was in line with the successful results reported in the previous literature, a more in-depth evaluation of the level achieved over time underline a more critical situation.

The level of high functional knee restoration needed for the high-demanding activity level of these patients can be challenging to achieve. This is confirmed by this series, where the joint status significantly influenced the final outcome. In fact, patients with a more compromised joint homeostasis, due to a pathological history requiring previous surgeries, presented a significantly lower rate of return to competitive sports. This reflects the importance of the joint status for the outcomes of cartilage procedures, as confirmed by other literature reports of degenerative conditions being correlated with worse outcomes [10, 11]. Degenerative etiology or patients previously treated with microfractures for the same defect [23] are considered as negative prognostic factors for MACT, and articular damages persisting for long time may irreversibly alter the joint environment and hamper the possibility to obtain optimal results. Saris et al. [23] confirmed that an altered intra-articular environment decreased the outcomes of tissue engineered cartilage repair in preclinical setting, and Ozsoy et al. [24] showed that an early treatment correlates with better results. Finally, Rodrigo et al. [28] showed that synovial fluid from patients with a fresh or old traumatic cartilage lesion can either stimulate or inhibit chondral healing depending on time from trauma. This was confirmed also in the present analysis since patients with degenerative/chronic lesions less likely returned to competitive sports compared to traumatic/acute ones. This should be taken into account when informing sport active patients with regard to their perspectives after treatment. Accordingly, OCD etiology was shown to be correlated to a low rate of return to sport. Although this population is usually characterized by a good prognosis [8, 9], and an overall positive clinical outcome was confirmed in this series, the need to obtain the more complex restoration of the entire osteochondral unit proved to be challenging, leading to a lower outcome in this high-demanding joints. This confirms previously reported findings underlying the limits of MACT in properly restoring the subchondral bone and the need for procedures targeted to the entire osteochondral unit to achieve optimal results [26].

Among other factors influencing the prognosis in this population of sport active patients, BMI was found to influence the outcomes in this series, where patients with lower BMI values had higher sports activity level at follow-up. Jaiswal et al.[15] previously found that overweight and obese patients had poorer functional outcomes. This is not the case of this population of sport active patients, but it may be still speculated that a heavier body may entail a higher stress to an already compromised joint. Age also represents a prognostic factor for return to activity following cartilage repair, as reported by several other authors [8, 21, 22]. This may be due to the effect of aging on body regenerative potential, as confirmed by the higher rate of adolescent patients returning back to sports with no limitations after knee surgery. It has to be underlined that young patients operated for the first time reached a very high rate of return to sport at short-term follow-up (almost 90%). On the other hand, the multivariate analysis can put age into a different perspective. Older athletes often present a longer sport activity history with more injuries and surgeries, and the presence of previous surgery was deemed detrimental in terms of prognosis, which suggest that, more than age, the joint status is the key for a successful outcome after cartilage surgery.

This study presents some limitations: the relatively low number of patients, the lack of a control group and no imaging evaluation to document the status of the regenerated tissue. Limited information is available in the literature on the effect of chondrocyte-based procedure on high-level athletes. However, the yearly prospective evaluation at long-term follow-up of this study allowed to document not only the rate of return to sport, but also the level achieved, the maintenance of the sport activity level over time, and factors affecting patients' prognosis in terms of return to sport.

The analysis highlighted an overall good clinical outcome maintained for up to 10 years. However, a significant reduction of the level of sport was observed after surgery and many patients, particularly those with more compromised joints that underwent previous surgery, did not return back to the same competitive level. In light of these results, care should be taken when giving indications for regenerative cartilage surgery in athletes, and the rates of return to sport according to the identified prognostic factors results should be considered to have realistic expectations when managing athletes who are candidate to cartilage procedures.

Conclusion

This study documented that, despite an overall satisfactory clinical improvement, only about 60% of athletes reached the highest goal of returning to sport at the same pre-injury level. Although small lesion size, young age, and strong motivation of this sport competitive patients could favor good results, up to 10 years results showed that the chondrocyte-based regenerative approach has some limitations in terms of sport-related outcomes. The level of high functional knee restoration needed for such high-demanding activity level can be challenging to achieve, especially in patients with a more compromised joint homeostasis. The return to sport rate varies significantly according to specific patient and lesion characteristics, such as previous surgery, BMI, etiology and age, which should be considered when treating athletes affected by cartilage lesions.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest. GF is consultant for: Finceramica Faenza S.p.A, Italy; Fidia farmaceutica, Italy; CartiHeal ltd (2009) Israel, EON Medica, Italy; and received Institutional Support from FinCeramica Faenza S.p.A.; Fidia farmaceutica, Italy, IGEA Biomedical, Italy, Zimmer-BIOMET, USA, Kensey Nash, USA.

Ethical approval The Hospital Ethics Committee and Internal Review Board approved this study.

Informed consent Informed consent of all patients was obtained.

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